

Exponential Rules Cheat Sheet

Complete exponent rules cheat sheet with product rule, quotient rule, power rule, zero exponent, negative exponents. Free PDF download.

Basic Exponent Rules

NAME	FORMULA	NOTES
Product Rule	$a^m \cdot a^n = a^{m+n}$	Same base: add exponents
Quotient Rule	$\frac{a^m}{a^n} = a^{m-n}$	Same base: subtract exponents
Power Rule	$(a^m)^n = a^{mn}$	Power of power: multiply exponents
Product to Power	$(ab)^n = a^n b^n$	Distribute exponent to product
Quotient to Power	$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$	Distribute exponent to quotient

Special Exponents

NAME	FORMULA	NOTES
Zero Exponent	$a^0 = 1$	Any nonzero base to 0 is 1
Exponent of 1	$a^1 = a$	
Negative Exponent	$a^{-n} = \frac{1}{a^n}$	Reciprocal
Negative in Denominator	$\frac{1}{a^{-n}} = a^n$	Move to numerator

Fractional Exponents

NAME	FORMULA	NOTES
Square Root	$a^{1/2} = \sqrt{a}$	
Cube Root	$a^{1/3} = \sqrt[3]{a}$	
nth Root	$a^{1/n} = \sqrt[n]{a}$	
General Fractional	$a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$	

Exponential Functions

NAME	FORMULA	NOTES
Natural Exponential	e^x	$e \approx 2.71828$
Exponential Growth	$A = A_0 e^{kt}$	$k > 0$ for growth
Exponential Decay	$A = A_0 e^{-kt}$	$k > 0$ for decay
Compound Interest	$A = P \left(1 + \frac{r}{n}\right)^{nt}$	n = compounding periods
Continuous Compounding	$A = Pe^{rt}$	Limit as $n \rightarrow \infty$

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